Zoey Zhu

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Education

Columbia University New York, NY
Bachelor of Arts in Computer Science - Mathematics, Statistics minor, GPA: 4.2 *Expected May 2029*

- **Relevant Coursework**: Advanced Programming, Data Structures and Algorithms, Discrete Math, Linear Algebra, Ordinary Differential Equations, Calculus III and IV, Probability Theory, Calc-based Intro to Statistics, Intermediate Microeconomics

Technical Skills

- **Programming Languages**: Python (numpy, pandas, matplotlib, healpy, astropy), Java, C, C#, HTML, CSS
- **Tools**: Git, GitHub, Unity, VS Code, Claude Code, Microsoft Office (Word, Excel, Powerpoint)
- **Languages**: English, Mandarin (fluent)

Experience

Recitation Leader Sept. 2026 – Present
Statistics Department, Columbia University *New York, NY*

- Co-lead weekly recitations for Calc-based Intro to Statistics by Prof. Cynthia Rush and assist students with course concepts and problem-solving

Research

Undergraduate Research Assistant May 2026 – Present
Astronomy Department, Columbia University *New York, NY*

- Built a Python pipeline to process 167 binary black hole skymaps, compute X-ray scattering and absorption optical depths, and identify candidate regions to search for X-ray counterparts
- Quantified and created visualizations for sky-region overlap between 90% GW event credible regions and favorable X-ray scattering zones using healpy and matplotlib
- Cleaned up photon data to remove point sources, supernova remnants, HII regions, and Gamma Ray Bursts before conducting search
- Designed and implemented search algorithm analyzing photon data to identify X-ray counterparts to binary black hole events

Undergraduate Research Assistant Sept 2026 – Present
Computer Science Department, Columbia University *New York, NY*

- Assist in developing and refining models to predict default risk for federally funded internet broadband programs
- Brainstorm and investigate potential risk signals for ISP defaults

Projects

LionSwipe | *HTML/CSS, JavaScript, Git* July 2026 – Present

- Built a full-stack web app for Columbia students to plan meals, track spending, and maximize usage of their dining plan
- Winning project for the Overcoming Obstacles category in the Stellic Pathfinders Challenge
- Selected for the summer 2026 cohort of Columbia Engineering's Start Me Up Bootcamp and presented value propositions, customer segment research, and a business model canvas
- Conducted 40 targeted customer discovery interviews before using research to motivate an idea pivot to target student dining
- Brainstormed and filmed promotional videos and designed flyers to promote the platform.

Poker Calculator | *Python, Git* July 2026 – Present

- Implemented a monte-carlo simulation to estimate EV of poker hands based on number of players (2-9), active hands, betting rounds (0-4), and table positions
- Developing an interactive CLI interface that recalculates EV as players act across betting rounds
- Designing EV-anchored bot opponents with configurable parameters such as assertiveness, looseness, and volatility

Zodoku | *Python, Flask, JavaScript, HTML/CSS, Git* Apr. 2026 – June 2026

- Built a full-stack Sudoku web app with a Flask backend and JavaScript frontend
- Implemented 10 strategies (eg. XY-wing) with step-by-step candidate visualization and strategy explanations
- Designed a puzzle generator with 4 difficulty levels and a 3-tier hint system that reveals cell location, strategy name, and solution step

Leadership

- **Columbia Undergraduate Math Society Treasurer** (July 2026 – Present): Manage club funds and coordinate weekly member dinner logistics and orders
- **Columbia Road Runners Treasurer** (July 2026 – Present): Write budget requests and oversee club funds for races and events
- **Columbia Application Development Initiative** (Sept. 2026 – Present): Reach out to companies and plan outreach events as part of the Internal/External committee
- **Columbia Club Golf Competitive Team** (Sept. 2026 – Present)